ROYAL EDGE — MATHEMATICAL METHODOLOGY

Formal Derivation of BPRI, BOWPRI, Leverage Models & Bayesian Shrinkage

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1. Match Leverage Pressure Index (P_t)

Match leverage quantifies the criticality of every ball based on win equity dynamics and resource scarcity across 867 matches (200,999 deliveries).

First Innings Formula:

```
P_1st = 1.0 + 0.40*(1 - B_t/120) + 0.35*(W_t/10) + 0.25*(R_t / (120 - B_t + 1))
```

Second Innings Formula:

```
P_2nd = 0.50*(RRR_t / CRR_req) + 0.35*(10 / (10 - W_t)) + 0.15*(120 / B_t)
```

Knockout Adjustment: A 1.15x multiplier is applied universally to playoff, qualifier, semi-final, and final encounters.

2. Empirical Bayesian Shrinkage Framework

To insulate the franchise against small sample volatility without eliminating emerging finishers or death specialists, metrics undergo shrinkage towards empirical priors:

Shrinkage Estimator: `theta_hat_i = [ n_i / (n_i + K) ] * x_bar_i + [ K / (n_i + K) ] * mu_0`

- **Batting Prior:** K = 60 balls faced, Prior Mean mu_0 = 128.50 SR.
- **Bowling Prior:** K = 50 legal balls, Prior Mean mu_0 = 8.45 Economy.

3. Batting Pressure Reliability Index (BPRI)

BPRI is a composite index scaled 0–100 evaluating performance under leverage (LEI >= 1.50):

BPRI = 0.35*HP_SR + 0.25*HP_Avg + 0.20*Death_SR(16-20) + 0.10*Chase_SR + 0.10*Delta_SR

4. Bowling Pressure Reliability Index (BOWPRI)

BOWPRI evaluates high-pressure containment and wicket breakthrough potency:

BOWPRI = 0.35*HP_Econ(inv) + 0.25*HP_Avg(inv) + 0.20*Death_Econ(inv) + 0.10*Wkts/Ball + 0.10*Delta_Econ(inv)

5. Uncertainty & Sample Saturation (SSSI)

Sample Size Saturation Index (SSSI %): Measures the percentage of empirical target sample volume acquired (60 balls batting / 50 balls bowling). It denotes data saturation, NOT a probability of future player success.

1,000-Iteration Bootstrap 95% Confidence Intervals: Reported for every metric. Overlapping intervals indicate that numerical differences may not represent statistically deterministic talent separation.